Integration Review

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1 Introductory Remarks

1.1 Purpose of Review

The purpose of this review is twofold. First, to test your understanding of the core integration techniques used in Differential Equations, and second, to act as a reference throughout your time in Differential Equations . Most of these problems exploit the core of the techniques of integration you'll use in DE. I encourage you to periodically work through this packet to assess and strengthen your integration skills during your time in Differential Equations. While Differential Equations can be very difficult, having a strong command of the fundamental integration techniques will make the subject significantly easier.

1.2 Difficulty of Problems

Most of these problems assume you have working knowledge of the derivatives and anti-derivatives of exponents, logs, trigonometric functions, and can easily utilize u-substitution.

1.3 Solutions

A corresponding solution manual is provided for this review. The manual includes only the essential steps for each problem; it is expected that you will fill in the intermediate algebraic details and recognize the methods implicitly used in the solutions.

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2 Integration by Parts

2.1 Overview

2.1.1 Derivation of IBP

$$y = uv$$

Take the derivative

$$\frac{dy}{dx} = \frac{d(vu)}{dx}$$

Now use the product rule

$$\frac{dy}{dx} = \frac{du}{dx}v + \frac{dv}{dx}u$$

Now integrate

$$\int \frac{dy}{dx}dx = \int \frac{du}{dx}vdx + \int \frac{dv}{dx}udx$$

Rearrange terms

$$\int \frac{du}{dx}vdx = \int \frac{dy}{dx}dx - \int \frac{dv}{dx}udx$$

Now implement the fundamental theorem of Calculus

$$\int \frac{du}{dx}vdx = y - \int \frac{dv}{dx}udx$$

Substitute y with uv

$$\int \frac{du}{dx}vdx = uv - \int \frac{dv}{dx}udx$$

You're now left with the classic definition of IBP

$$\int u'vdx = uv - \int v'udx$$

2.1.2 Example

Problem:

$$\int xcos(x)dx$$

Solution:

$$\int u'vdx = uv - \int uv'dx$$

Decide u' and v:

$$u' = \cos(x) \quad \text{and} \quad v = x$$

Compute derivative of v and anti-derivative of u':

$$u = sin(x) \quad \text{and} \quad v' = 1$$

Now plug into IBP equation:

$$\int xcos(x)dx = xsin(x) - \int (1)(sin(x))dx$$

Compute integral:

$$\int xcos(x)dx = xsin(x) + cos(x) + C$$

Your final answer is:

$$xsin(x) + cos(x) + C$$

2.2 Problem 1

$$\int xln(x)dx$$

2.3 Problem 2

$$\int x^2e^xdx$$

2.4 Problem 3

$$\int x^2sin(x)dx$$

2.5 Problem 4

$$\int e^xsin(x)dx$$

2.6 Problem 5

$$\int cos(ln(x))dx$$

3 Partial Fractions

3.1 Overview

3.1.1 Explanation of Method

If some rational function:

$$\frac{P(x)}{Q(x)}, \quad \deg P(x) < \deg Q(x)$$

then you can factor Q(x) then decompose based on the factor types.

Factor Types:

Linear Factors:

$$Q(x) = (x-a)^n$$

,

then

$$\frac{P(x)}{Q(x)} = \frac{P(x)}{(x-a)^n} = \frac{A_1}{(x-a)} + \frac{A_2}{(x-a)^2} + ... + \frac{A_n}{(x-a)^n}$$

Quadratic Factors:

$$Q(x) = (ax^2 + bx + c)^n$$

then

$$\frac{P(x)}{Q(x)} = \frac{P(x)}{(ax^2+bx+c)^n} = \frac{A_1}{(ax^2+bx+c)} + \frac{A_2}{(ax^2+bx+c)^2} + ... + \frac{A_n}{(ax^2+bx+c)^n}$$

3.1.2 Example

Problem:

$$\int \frac{x^3+2}{(x-1)^2(x^2+1)} dx$$

Solution: Decompose integral into respective factors

$$\int \frac{x^3+2}{(x-1)^2(x^2+1)} dx = \int \frac{A}{x-1} + \frac{B}{(x-1)^2} + \frac{Cx+D}{(x^2+4)} dx$$

Now multiply across with the factors and solve for the constants

$$x^3 + 2 = [\frac{A}{x-1} + \frac{B}{(x-1)^2} + \frac{Cx+D}{(x^2+4)}][(x-1)^2(x^2+1)]$$

$$x^3 + 2 = A(x-1)(x^2+1) + B(x^2+1) + (Cx+D)(x^2-1)$$

$$(x-1)^2 = x^2 - 2x + 1 \quad \text{and} \quad (x^2+1)(x-1)^2 = x^4 - 2x^3 + 2x^2 - 2x + 1$$

$$x^3 + 2 = A(x^3 - x^2 + x - 1) + B(x^2+1) + C(x^3 - 2x^2 + x) + D(x^2 - 2x + 1)$$

$$x^3 + 2 = x^3(A+C) + x^2(-A+B-2C+D) + x(A+C-2D) + (-A+B+D)$$

Now solve using your preferred algebraic method

$$A = 0 \quad B = \frac{3}{2} \quad C = 1 \quad D = \frac{1}{2}$$

Now plug constants into integral and solve

$$\int \frac{0}{x-1} + \frac{3/2}{(x-1)^2} + \frac{x+(1/2)}{(x^2+1)} dx$$

$$\int \frac{3}{2(x-1)^2} + \frac{x}{x^2+1} + \frac{1}{2(x^2+1)} dx$$

Final Answer

$$\frac{-3}{2(x-1)} + ln|(x^2+1)^{1/2}| + \frac{1}{2} tan^{-1}(\frac{x}{2}) + C$$

3.2 Problem 6

$$\int \frac{1}{x^2(x^2+1)}dx$$

3.3 Problem 7

$$\int \frac{3x+2}{(x-2)^2(x^2+1)}dx$$

3.4 Problem 8

FYI: After working through this problem, I realized that it is comparatively more difficult than the others and may not be worth the time required to solve it.

$$\int \frac{x-3}{x(x^3-27)}dx$$

3.5 Problem 9

$$\int \frac{4}{x^2+5x-14}dx$$

3.6 Problem 10

$$\int \frac{4x-11}{x^3-9x^2}dx$$

4 Trigonometric Techniques

4.1 Overview

4.1.1 Quick Remarks

Most trigonometric integrals encountered in Differential Equations are evaluated using trigonometric identities or the standard anti-derivative formulas for the basic trigonometric functions. Listed below are several identities and anti-derivative formulas you are expected to be familiar with, but which may be useful to review.

4.1.2 Trigonometric Identities

Addition Formulas

$$sin(A+B) = sin(A)cos(B) + cos(A)sin(B)$$

$$cos(A+B) = cos(A)cos(B) - sin(A)sin(B)$$

Euler's Formulas

$$e^{ix} = cos(x) + isin(x)$$
$$e^{-ix} = cos(x) - isin(x)$$

Pythagorean(important)

$$sin^2(x) + cos^2(x) = 1, \quad 1 + tan^2(x) = sec^2(x), \quad 1 + cot^2(x) = csc^2(x)$$

Power-Reduction Formulas(important)

$$sin^2(x) = \frac{1 - cos(2x)}{2}, \quad cos^2(x) = \frac{1 + cos(2x)}{2}$$

Inverse Trig Integrals

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx == sin^{-1}(\frac{x}{a}) + C$$

$$\int \frac{1}{x^2 + a^2} dx = \frac{1}{a} tan^{-1}(\frac{x}{a}) + C$$

$$\int \frac{a}{x\sqrt{x^2 - a^2}} dx = \frac{1}{a} sec^{-1}(\frac{x}{a}) + C$$

4.2 Problem 11

$$\int 2sin^2(7x)dx$$

4.3 Problem 12

$$\int \frac{5}{13} cos^2(12x)dx$$

4.4 Problem 13

$$\int cos^2(x)sin^3(x)dx$$

4.5 Problem 14

$$\int \frac{2}{8 + x^3} dx$$

4.6 Problem 15

$$\int \frac{14}{\sqrt{7 - x}} dx$$

4.7 Problem 16

$$\int tan(x)dx$$

4.8 Problem 17

$$\int sin^3(x)dx$$

4.9 Problem 18

$$\int e^{7+cos(x)}sin(x)dx$$

5 Logarithmic/Exponential Techniques

5.1 Overview

5.1.1 Formulas

Assume a is some constant

$$\int e^{ax}dx = \frac{e^{ax}}{a} + C, \quad \int \frac{a}{x}dx = ln|x^a| + C$$

5.2 Problem 19

$$\int \frac{e^{1/x}}{x^2}dx$$

5.3 Problem 20

$$\int \frac{e^{ln(2x)}}{3x}$$

5.4 Problem 21

$$\int ln(cos(x))tan(x)dx$$

5.5 Problem 22

$$\int \frac{ln(x)(3+ln^2(x))^{1/2}}{x}dx$$

5.6 Problem 23

$$\int sec^2(x)cot(x)dx$$

6 Algebraic Techniques

6.1 Overview

6.1.1 Completing the Square

Assume you have a quadratic in standard form that is not easily factored. You can use complete the square to turn the quadratic from standard form, to vertex form.

$$ax^2 + bx + c = a(x-h)^2 + k$$

$$h = \frac{-b}{2}, \quad k = c - h^2 = c - \frac{b^2}{4}$$

6.1.2 Synthetic Division

If you are not already familiar with synthetic division, I strongly encourage you to watch a video tutorial on the topic before attempting the problems below. Although the procedure itself is straightforward, it is unlikely to become intuitive from a brief in-text explanation alone.

6.2 Problem 24

$$\int \frac{1}{x^2 + 3x + 3} dx$$

6.3 Problem 25

$$\int \frac{1}{x^2 - 2x + 23} dx$$

6.4 Problem 26

$$\int \frac{2x^3 + 3x^2 - 4x + 8}{(x+3)} dx$$

6.5 Problem 27

$$\int \frac{x^4 + x^2}{x - 2} dx$$

6.6 Problem 28

$$\int \frac{x^3 - 1}{x - 1}$$

7 Misc. Problems

7.1 Remarks

If you successfully solved most of the problems above, you likely have a good understanding of the necessary integral techniques to be successful in DE. The purpose for the problems below are purely supplemental, and are provided for those who want extra review.

7.2 Problem 29

$$\int 16xsin(13+7x)dx$$

7.3 Problem 30

$$\int e^{3x}cos(\frac{x}{7})dx$$

7.4 Problem 31

$$\int x^3cos(2x^4)dx$$

7.5 Problem 32

$$\int (3x^3+2x^2-7x)(e^x)dx$$

7.6 Problem 33

$$\int sex^4(2x)tan^2(2x)dx$$

7.7 Problem 34

$$\int \frac{1+7sin^3(x)}{cos^2(x)}dx$$

7.8 Problem 35

$$\int \frac{8}{3x^3+7x^2+4x}$$

7.9 Problem 36

$$\int \frac{8x}{x^3-x^2}dx$$

7.10 Problem 37

$$\int \frac{1}{e^x\sqrt{e^{2x}-9}}dx$$

7.11 Problem 38

$$\int (\sqrt{x}-x^{-1})^2 dx$$

7.12 Problem 39

$$\int \frac{e^{\sqrt{3}}+x^{\sqrt{2}}}{x^{1/2}}dx$$

7.13 Problem 40

$$\int \frac{e^{x/3}}{(34-e^{x/3})^{5/2}}dx$$

7.14 Problem 41

$$\int tan(x)ln(cos(x))dx$$

7.15 Problem 42

$$\int \sqrt{25-x^2}dx$$

7.16 Problem 43

$$\int ln^2(x)dx$$

7.17 Problem 44

$$\int \frac{ln(x)}{x^4}dx$$

7.18 Problem 45

$$\int e^{6x}sin(e^{6x})dx$$

7.19 Problem 46

$$\int \frac{10}{(x-1)(x^2+9)}dx$$